

LEADER & ENTHUSIAST COURSE [PHASE-ALL]

TARGET : PRE-MEDICAL 2020

Test Type : Full Syllabus

Test # 03

Test Pattern : NEET-UG

TEST DATE : 30 - 04 - 2020

ANSWER KEY

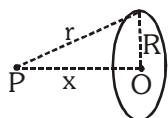
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	3	4	1	1	3	2	3	1	1	3	4	1	3	4	3	4	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	3	2	2	2	2	4	1	4	3	2	3	1	3	3	3	1	3	3	2
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	2	4	2	4	1	1	3	2	1	4	3	2	4	3	2	1	2	2
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	3	1	1	2	2	1	4	2	3	2	4	3	2	4	3	1	3	4	2	3
Que.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Ans.	2	3	3	4	2	4	4	3	4	2	3	4	4	4	1	3	3	3	2	1
Que.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	2	3	2	4	1	4	2	4	3	4	3	4	1	4	3	3	3	4	4	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
Ans.	3	4	3	4	3	4	3	2	4	4	3	4	3	2	2	2	2	3	4	2
Que.	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160
Ans.	2	3	3	2	3	2	3	2	1	4	3	3	3	2	4	3	2	4	2	2
Que.	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	1	3	2	1	1	3	3	3	3	1	1	2	4	3	4	1	4	2	3	2

HINT - SHEET

1. $E = \frac{kQx}{(R^2 + x^2)^{3/2}}$

$$r^2 = R^2 + x^2$$

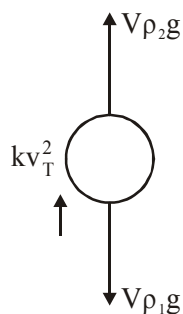
$$x^2 = r^2 - R^2$$



2. The force acting on the ball are gravity force, buoyancy force and viscous force. When ball acquires terminal speed, it is in dynamic equilibrium, let terminal speed of ball is v_T

$$\text{So, } V\rho_2g + kv_T^2 = V\rho_1g$$

$$v_T = \sqrt{\frac{V(\rho_1 - \rho_2)g}{k}}$$



3. $r = \frac{\sqrt{2mE_k}}{qB}$

$$r \propto \sqrt{m} \quad (q, B, E_k \rightarrow \text{same})$$

$$\text{or } m \propto r^2$$

$$\frac{m_1}{m_2} = \left(\frac{r_1}{r_2}\right)^2$$

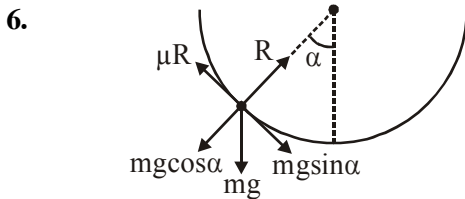
4. The coin will have no motion relative to the gramophone record when inertial force $mR\omega^2$ is less than or equal to the static frictional force μmg .

$$\therefore mR\omega^2 \leq \mu mg$$

$$\text{or } R \leq \frac{\mu g}{\omega^2}$$

5. $\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda} \quad \{\Delta x = x\}$

$$\Delta\phi = \frac{2\pi x}{\lambda}$$



$$\mu R = mg \sin \alpha$$

$$\text{or } \mu mg \cos \alpha = mg \sin \alpha$$

$$\cot \alpha = \frac{1}{\mu} = 3$$

7. $V_2 = V_3$

$$Q_1 = Q_2 + Q_3$$

$$V = V_1 + V_2$$

8. Let ρ be the density of lead

$$\text{Then, } M = \frac{4}{3}\pi R^3 \rho = \text{mass of total sphere}$$

$$m_1 = \frac{4}{3}\pi \left(\frac{R}{2}\right)^3 \rho = \text{mass of removed part} = \frac{M}{8}$$

$$m_2 = M - \frac{M}{8} = \frac{7M}{8} = \text{mass of remaining sphere}$$

Choosing the centre of big sphere as the origin,

$$X_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$0 = \frac{\frac{M}{8} \times \frac{R}{2} + \frac{7M}{8} \times x_2}{M}$$

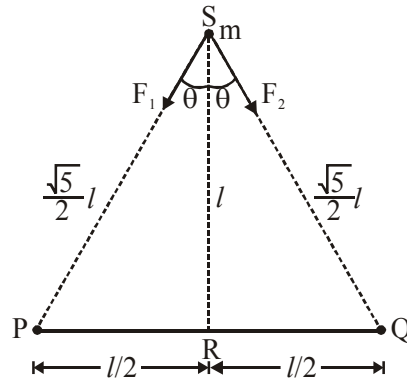
$$\text{Solving, we get ; } x_2 = \frac{-R}{14}$$

i.e., centre of mass of hollowed sphere be at a distance of $R/14$ on left on O,

$$\text{i.e., shift in centre of mass} = \frac{R}{14}.$$

11. $T_2 = T_1 + 4 \times 9.8$
 $6 \times 9.8 = T_1 + 4 \times 9.8$
 $T_1 = 2 \times 9.8$
 $T_1 = 19.8 \text{ N}$

12. $F_1 = \frac{4Gm^2}{5l^2} = F_2$



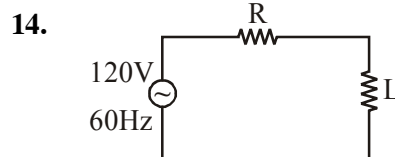
$$\cos \theta = \frac{2l}{\sqrt{5}l} = \frac{2}{\sqrt{5}}$$

Resultant force at S is

$$F = F_1 \cos \theta + F_2 \cos \theta = 2 F_1 \cos \theta$$

$$= \frac{8Gm^2}{5l^2} \cdot \frac{2}{\sqrt{5}} = \frac{16Gm^2}{5\sqrt{5}l^2}$$

13. In above question to get COM at distance maximum from centre, distance should be $\frac{r}{14}$



$$V_R^2 + V_L^2 = 120^2$$

$$V_L = \sqrt{120^2 - 54^2} = 107 \text{ V}$$

$$\therefore V_R i = P \quad i = \frac{16}{54} \text{ A}$$

$$\text{Now } i 2\pi f L = V_L$$

$$\Rightarrow \frac{16}{54} \times 2 \times \pi \times 60 L = 107$$

$$\Rightarrow L = 0.96 \text{ H} \approx 1 \text{ H}$$

16. $T_2 - T_1 = M_2 a \quad \dots(i)$

$$T_1 = M_1 a \quad \dots(ii)$$

$$\therefore T_2 - T_1 = M_2 \times \frac{T_1}{M_1}$$

$$\text{or } T_2 = T_1 \left[1 + \frac{M_2}{M_1} \right] \quad \text{or } \frac{T_1}{T_2} = \frac{M_1}{(M_1 + M_2)}$$

17. $T_1 = \frac{12 \times 10}{1W} ACW$

$T_2 = \frac{5 \times 1}{2} \times \frac{10}{1W} ACW$

$T_3 = 0$

$T_4 = \frac{4 \times 10}{1W} CW$

$T_{net} = \frac{12 \times 10}{1W} + \frac{5 \times 10}{2W} - \frac{4 \times 10}{1W} NM$

$= 1.2 + 0.25 - 0.4 ACW$

$= 1.05 NM ACW$

18. $B_{out} = \frac{\mu_0 i}{2\pi(2a)} \dots (1)$

$B_{in} = \frac{\mu_0 i(a/2)}{2\pi(a)^2} \dots (2)$

From (1) & (2)

$\frac{B_{out}}{B_{in}} = \frac{\mu_0 i / 4\pi a}{\mu_0 i / 4\pi a} = \frac{1}{1}$

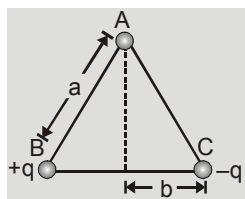
19. $E_y = 2.5 \frac{N}{C} \cos[(2\pi \times 10^6 t) - \pi \times 10^{-2} x]$

direction - X axis

wavelength $\lambda = \frac{2\pi}{k} = \frac{2\pi}{\pi \times 10^{-2}} = 200 \text{ m}$

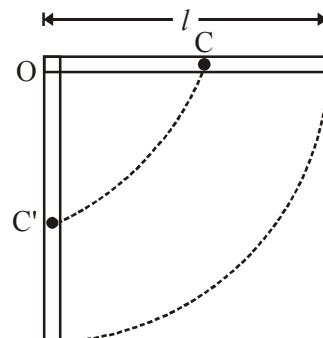
$f = \frac{\omega}{2\pi} = \frac{2\pi \times 10^6}{2\pi} = 10^6 \text{ Hz}$

21.



$V_A = \frac{k(+q)}{a} + \frac{k(-q)}{a} = 0$

22. Mass of the rod is concentrated at its centre of mass. In the vertical position of the rod, the centre of mass is at C'. Now loss in PE = Gain in Rotational KE



$mg \frac{l}{2} = \frac{1}{2} I \omega^2$

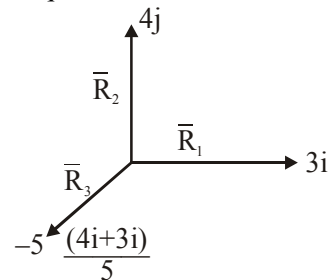
Where $I = \frac{ml^2}{3}$

The moment of inertia of the rod about axis passing through O and 1 to the plane of paper

$\frac{mg\ell}{2} = \frac{1}{2} = \frac{m\ell^2}{3} \omega^2$

$\omega = \frac{\sqrt{3g}}{\ell}$

23. In a plane to produce zero resultant, minimum vector required are 3



24. $\sin \theta_c = \frac{1}{\mu}; \sin 45^\circ = \frac{1}{\mu}$
 $\mu = \sqrt{2}$

25. $R = \lambda \cdot N = \lambda \frac{m}{M_w} \times N_A$

26. $\frac{R_1 R_2}{R_1 + R_2} = \frac{8}{3}$ & $R_1 + R_2 = 12 \Rightarrow R_1 R_2 = 32$

$\Rightarrow R_2 - R_1 = \sqrt{(R_1 + R_2)^2 - 4R_1 R_2}$

$= \sqrt{12^2 - 4 \times 32} = 4\Omega$

So $R_1 = 4\Omega$ and $R_2 = 8\Omega$

Hence $\frac{\ell_1}{\ell_2} = \frac{4}{8} = \frac{1}{2}$

27. In steady state condition capacitor behave as open circuit.

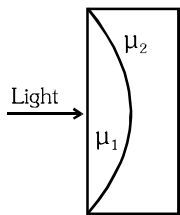
28. $t = \alpha x^2 + \beta x = x(\alpha x + \beta)$

$$1 = 2\alpha \frac{dx}{dt} \cdot x + \beta \frac{dx}{dt}$$

$\therefore$

$$v = \frac{dx}{dt} = \frac{1}{\beta + 2\alpha x}; \quad \frac{dx}{dt} = \frac{-2\alpha v}{(\beta + 2\alpha x)^2} = 2\alpha v^3$$

29.

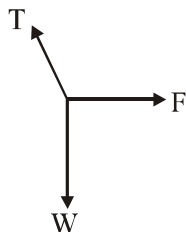


Equivalent focal length is given by $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$

$$\frac{1}{f_{eq}} = (\mu_1 - 1) \left(\frac{1}{\infty} - \frac{1}{-R} \right) + (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

$$\Rightarrow f_{eq} = \frac{R}{\mu_1 - \mu_2}$$

30.



(observer is on moving frame so centrifugal force act outward)

31.



$$R_{eq} = R_1 + R_2$$

$$\Rightarrow \frac{2\ell}{\sigma_{eq} A} = \frac{\ell}{\sigma_1 A} + \frac{\ell}{\sigma_2 A} \Rightarrow \sigma_{eq} = \frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$$

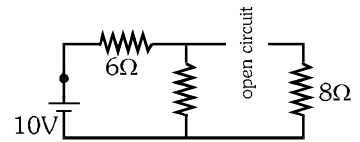
32. According to principle of conservation of angular momentum, we get ;

$$I_1 \omega_1 = I_2 \omega_2$$

$$\frac{2}{5} MR^2 \times \frac{2\pi}{24} = \frac{2}{5} M \left(\frac{R}{n} \right)^2 \frac{2\pi}{T'}$$

$$\therefore T' = \frac{24}{n^2} \text{ hours}$$

33. At $t = 0$



$$I = \frac{V}{R_{net}} = \frac{10}{6 + 8} = \frac{10}{14} = \frac{5}{7} \text{ A}$$

34. The frequency of vibrations of a sonometer wire is given by :

$$n = \frac{1}{2L} \sqrt{\frac{T}{m}}$$

The tuning fork produces 5 beats per second with lengths 20 cm and 21 cm. If n be frequency of fork, then

$$n + 5 = \frac{1}{2 \times 0.20} \sqrt{\frac{T}{m}} \quad \dots(i)$$

$$n - 5 = \frac{1}{2 \times 0.21} \sqrt{\frac{T}{m}} \quad \dots(ii)$$

Dividing eqn. (i) by (ii) we get :

$$\text{hence, } \frac{n + 5}{n - 5} = \frac{0.21}{0.20}$$

$$\begin{aligned} \text{or } 0.2n + 1 &= 0.21n - 1.05 \\ \text{or } 1 + 1.05 &= 0.21n - 0.2n \\ \text{or } 0.05 &= 0.01n \\ \text{or } n &= 205 \text{ Hz} \end{aligned}$$

36. Heat absorbed in a thermodynamic process is given by

$$\Delta Q = \Delta U + \Delta W$$

Here, ΔU is same for all the three processes as it depends only on initial and final states.

$$\text{But } \Delta W_I = +ve, \Delta W_{II} = 0$$

$$\text{and } \Delta W_{III} = -ve$$

$$\text{or } \Delta Q_I > \Delta Q_{II}$$

37. As $a = \frac{dv}{dt} = \frac{F}{m} = \text{constant}$, hence, on integration,

we get ;

$$v = \frac{F}{m} t$$

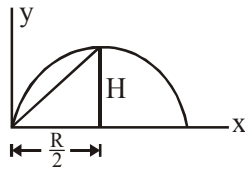
$$\text{So, power, } P = Fv = \frac{F^2}{m} t$$

$$\text{i.e., } p \propto t$$

For power to be constant, $Fv = \text{constant}$, i.e.,

$$F \propto \frac{1}{v}$$

38. Average velocity = $\frac{\text{Displacement}}{\text{Time}}$



$$v_{av} = \frac{\sqrt{H^2 + \frac{R^2}{4}}}{T/2}$$

Here, H = maximum height

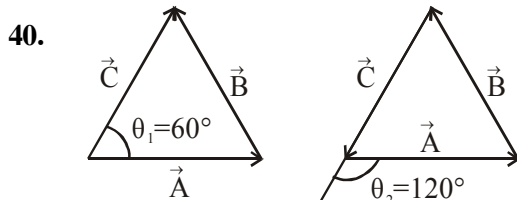
$$= \frac{v^2 \sin^2 \theta}{2g}$$

$$R = \text{range} = \frac{v^2 \sin 2\theta}{g}$$

and T = time of flight = $\frac{2v \sin \theta}{g}$

$$\therefore v_{av} = \frac{v}{2} \sqrt{1 + 3 \cos^2 \theta}$$

39. $f^1 = \frac{340 + 20}{340 - 20} \times 480 = 540 \text{ Hz}$



41. W = Area of rectangle
 $= -100 \times 10^3 \times 200 \times 10^{-6}$
 $= -20 \text{ J}$

-ve sign is taken because the cycle is anticlockwise.

42. Power $P = Fv = \frac{K}{v} \cdot v = K = \text{constant}$

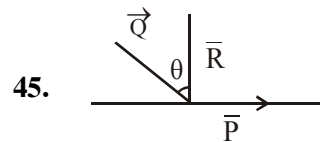
$$\therefore W = Pt = Kt$$

43. Induced emf will not remain constant in circular and elliptical loops.

44. $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

$$I_R = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi$$

$$I_R = 2I_0(1 + \cos \phi)$$



45.

$$\text{Now } Q \sin \theta = P$$

$$\& \quad Q \cos \theta = 2P$$

$$\tan \theta = 1/2 \quad \text{so, } \sin \theta = 1/\sqrt{5}$$

$$P = 8(1/\sqrt{5}) = \frac{8}{\sqrt{5}}$$

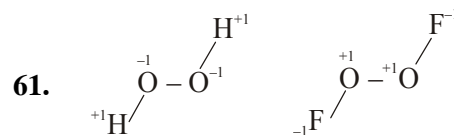
46. On moving half to light in a periodic table

$Z_{eff} \uparrow$ Size $\downarrow$

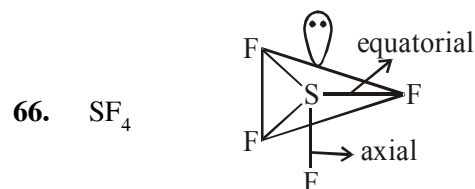
51. Due O to H-bonding

56. $\text{XeO}_3 = sp^3$ %s Character = 25%
 More EN

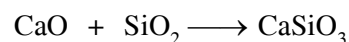
57. Pseudohalide contain nitrogen.



62. H_3PO_3 Hydrogen atoms = 3
 (फॉस्फोरस अम्ल)

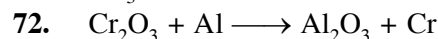


67. In extraction of iron SiO_2 is impurity so use basic flux.



Flux Impurity Slag

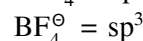
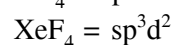
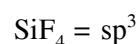
71. NF_3 central atom does not have vacant orbital.



Al work as reducing agent.

76. $\text{SF}_4 = sp^3d \Rightarrow$ Bond length order

Axial > equatorial



81. ClO_2 is paramagnetic nature due to add e^- species.

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| <p>93. NCERT (XII) Pg # 79</p> <p>95. Module No.6, Page # 125</p> <p>97. NCERT XIth Pg.#163</p> <p>103. NCERT (XII) Pg # 209</p> <p>104. NCERT XI, Page # 128</p> <p>105. Module No.6, Page # 58</p> <p>107. NCERT XIth Pg.#133</p> <p>113. NCERT (XII) Pg # 202</p> <p>115. Module No.6, Page # 60</p> <p>120. NCERT XIth Pg. # 308 Fig. - 20.5</p> <p>123. NCERT (XII) Pg # 212</p> <p>125. Module No.6, Page # 61</p> <p>130. NCERT XIth Pg. # 311 Fig. - 20.7</p> <p>131. NCERT Pg # 88</p> <p>133. NCERT XI, Page # 30</p> | <p>135. Module No.6, Page # 59</p> <p>136. NCERT XII Pg.# 177 (E), 193 (H)</p> <p>140. NCERT XIth Pg. # 319 Fig. - 21.3</p> <p>142. NCERT (XII) Pg # 117</p> <p>143. NCERT XI, Page # 8</p> <p>145. Module No.6, Page # 92</p> <p>146. NCERT XII Pg.# 174 (E), 189 (H)</p> <p>147. NCERT XI Pg.# 265. Table 16.1</p> <p>151. NCERT Pg # 76</p> <p>157. NCERT XI Pg.# 259</p> <p>164. Module No.6, Page # 07</p> <p>172. NCERT (XII) Pg # 79</p> <p>173. NCERT XI, Page # 38</p> <p>174. Module No.6, Page # 125</p> <p>178. Module - 6 Page # 65</p> |
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